

LABORATÓRIO DE GRANDES DADOS

Laboratório de Grandes Dados
Mestrado em Engenharia Informática
Engenharia de Dados
ISEP/P.PORTO

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TOPICS

- Data
- Data Literacy
- Data-Driven
- AI Literacy
- AI Leadership

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DATA TRENDS IN 2025

1. Automated Insights Become Universal
2. Synthetic Data Takes Center Stage
3. Confidential Computing Emerges As A Game-Changer
4. Real-Time Data Becomes Non-Negotiable
5. Data Sovereignty Sparks Global Debate
6. Data Trading Markets Explode
7. Data-Centric AI Revolutionizes Machine Learning
8. Regulatory Frameworks Take Shape
9. Edge Computing Transforms Data Processing
10. Behavioral Data Becomes The New Gold

The 10 Most Powerful Data Trends
That Will Transform Business In 2025

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Data Revolution ...

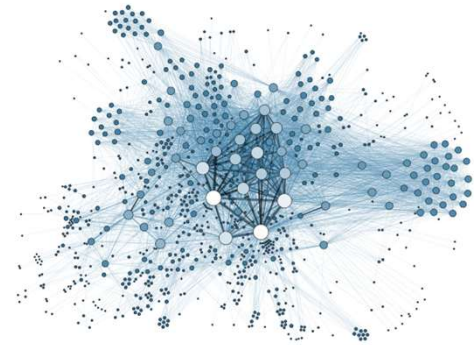
- According to IBM research, 90% of the world's data was created during the last two years,
- in the 2030s, it's estimated that we will be generating a Yottabyte of data per year....
- From bytes to yottabytes: **the data revolution**
- **Data-based decision** making has emerged and evolved significantly
- **New data is complex and unstructured!**



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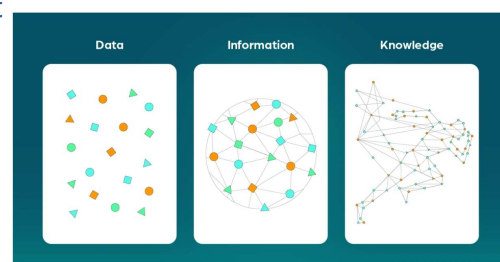
Data is everywhere...

- ... so much information in so many different forms ...
- today ... significantly more data exists, and it varies in type and timeliness
- Need to managing data differently!
- **Data management** has to include technology advances in hardware, storage, networking, and computing models (such as virtualization and cloud computing).
- The convergence of **emerging technologies** and **reduction in costs** have transformed the data landscape and made **new opportunities** possible.

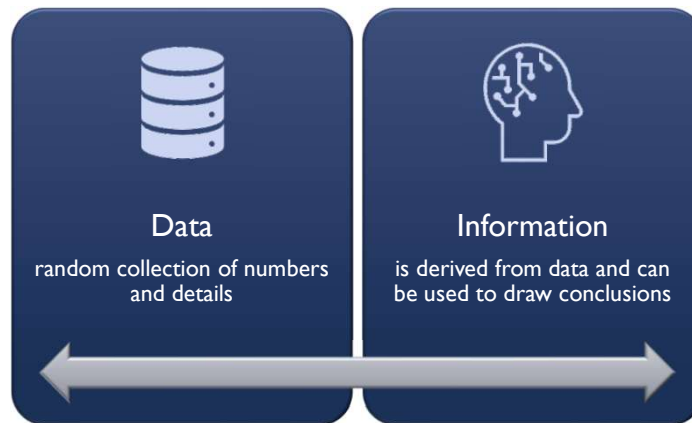


Data vs Information vs Knowledge

- **Data** is the starting point of knowledge management and can be best linked to an organization's raw materials.
- **Information** is a collection of data that has been processed, refined, structured, and/or presented to create relevance.
- **Knowledge** makes meaning from information, based on experience and intuition.



DATA IS EVERYWHERE...



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DATA LITERACY VS AI LITERACY

- Upskilling & Reskilling in the AI Era
- where does data literacy end and AI literacy begin? T
- AI literacy as a natural extension of data literacy
- Understanding the intricacies of generative AI requires strong data literacy
- AI will lower the barrier to data literacy
- The launch of ChatGPT made AI accessible to the world.
- AI and data literacy are complimentary skill sets that supports data-driven and analytical world.

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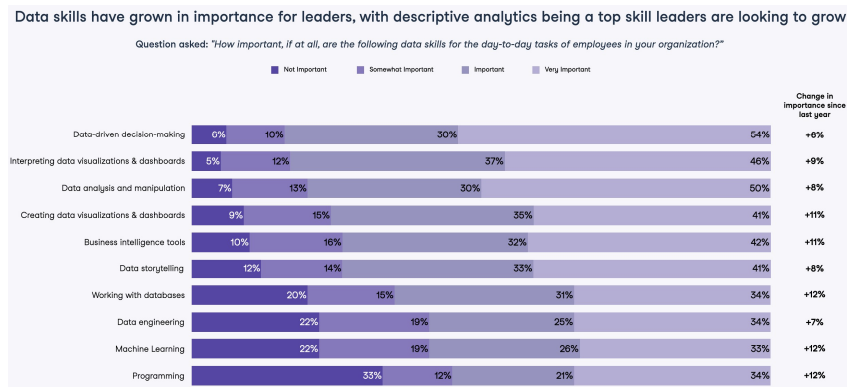
DATA LITERACY VS AI LITERACY

- **The evolving data & AI skill gap**
- **AI literacy** means understanding how AI works in the context of their work.
 - What is relevant to the chief marketing officer may not be the same for human resources, accounting, or finance.
- **Data Literacy** as the ability to read, write, analyze, communicate, and argue with data
- **AI Literacy** as the ability to effectively, ethically, and responsibly understand, use and guide AI system

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THE STATE OF DATA & AI LITERACY 2024



Source: The state of data and AI literacy in 2024, Datacamp

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MAIN CHALLENGES TO HUMANITY

- Loss of jobs is not the real risk of AI and technology developments
- The threat is not AI, but people using AI
- Based on study of World Economic Forum - Global Risks Perception Survey (2024):
 - **misinformation and disinformation** represent the most significant risk facing humanity in the immediate future
 - widen societal and political divides

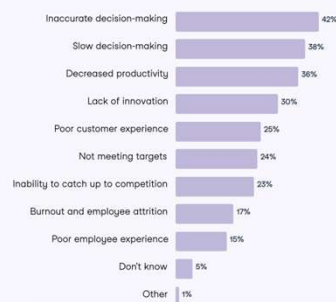
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THE IMPACT OF DATA SKILLS

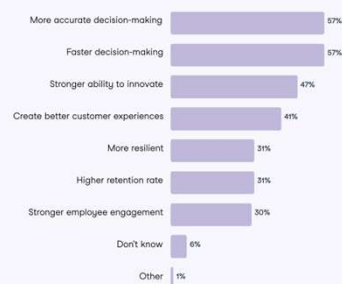
Leaders face a variety of risks if they do not build their workforce's data skills

Question asked: "What risks is your department or team facing if your people do not have adequate data skills?"



The value placed on existing employees with adequate data skills shows a similar story

Question asked: "What value do data-literate employees provide over those with insufficient data skills?"



Source: The state of data and AI literacy in 2024, Datacamp

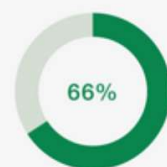
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MAIN RISKS TO HUMANNITY

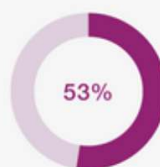
Risk categories

- Economic
- Environmental
- Geopolitical
- Societal
- Technological



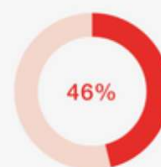
1st

Extreme weather



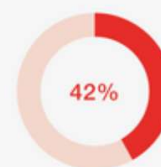
2nd

AI-generated misinformation and disinformation



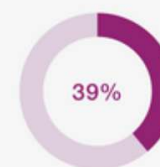
3rd

Societal and/or political polarization



4th

Cost-of-living crisis



5th

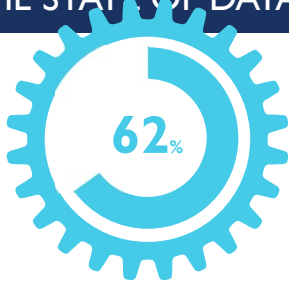
Cyberattacks

Source: Global Risks 2024 Report - World Economic Forum

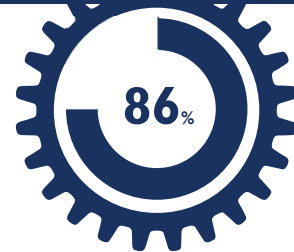
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THE STATE OF DATA & AI LITERACY 2024



62% of leaders believe AI literacy is important for their teams' daily tasks



86% of leaders believe data literacy is important for their teams' daily tasks. Up 3% since last year.

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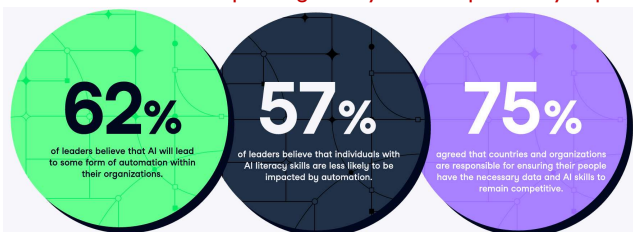
Source: The state of data and AI literacy in 2024, Datacamp

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AI IMPACT BEYOND THE WORKPLACE

- majority of jobs will be augmented
- AI will make us better and more efficient and hopefully more human in the long run.

"... four in 10 workers expressing anxiety about AI potentially displacing their roles"

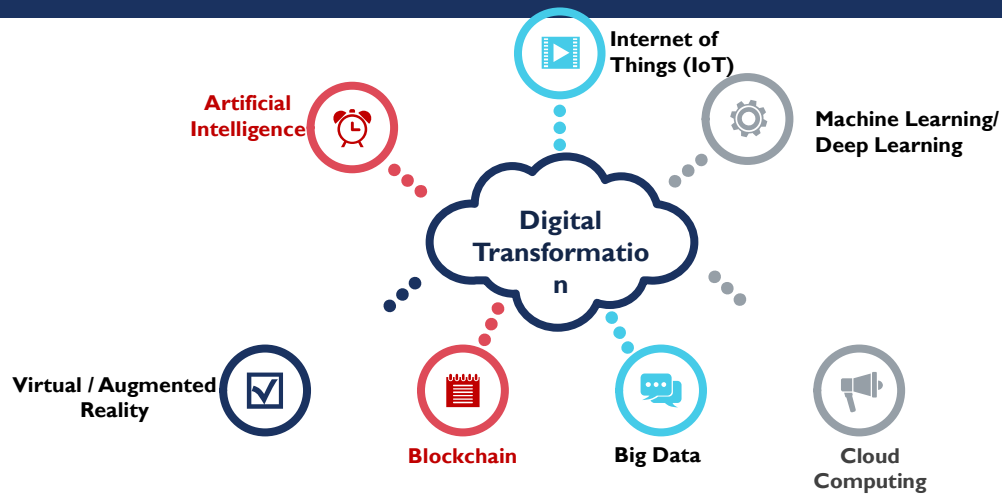


AI literacy as a safeguard against the potential impact of generative AI on job security

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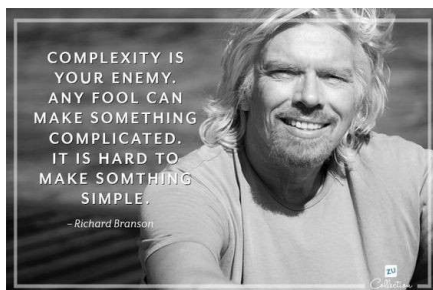
WITH DIGITAL TRANSFORMATION



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ARTIFICIAL INTELLIGENCE



AI is not just a tool:

- it is an extension of human ingenuity and creativity.
- It is about leveraging AI to amplify our strengths,
- address global challenges,
- and create a more equitable and sustainable future

Requires a deep understanding of AI capabilities and limitations.

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FROM DATA TO BIG DATA

- **Big Data**
- **Data Science/Data Engineering/Data Analytics**

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FROM DATA TO BIG DATA

- **From bytes to yottabytes:** the data revolution
- **Data-based decision making** has emerged and evolved significantly



The availability of massive amounts of good data and fast computation



New data is complex and unstructured

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DATA-DRIVEN

- the practice of **collecting and analyzing data to derive insights and solutions.**
- A **data-driven approach** helps us predict the future using past and current data.
- Without data, we run the risk of making false assumptions and being influenced by biased conclusions.

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WHAT IS BIG DATA?

- “Big data is important because it enables organizations to gather, store, manage, and manipulate vast amount of data at the right speed, at the right time, to gain the right insights.”
- **is not a stand-alone technology** ... it is a combination of the last 50 years of technology evolution.
- ...as any kind of data source that has at least three shared characteristics:
 - ✓ Extremely large Volumes of data
 - ✓ Extremely high Velocity of data
 - ✓ Extremely wide Variety of data
- Big data refers to extremely large and diverse data sets that are not easily managed with traditional data processing methods and tools.
- Big data technology is changing at a rapid steps....

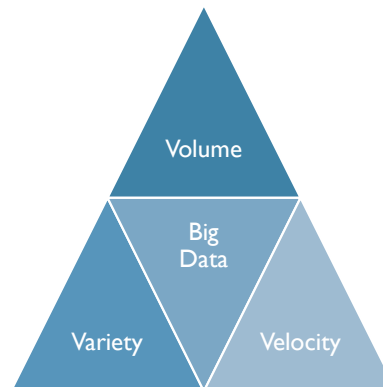


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DATA SCIENCE, BIG DATA, AND DATA ANALYTICS IN HEALTHCARE

Big Data refers to the vast and complex data sets that cannot be easily managed or processed using traditional data processing tools. The characteristics of Big Data are often referred to as the three V's:

- **Volume**—How much data is there?
- **Variety**—How diverse are different types of data?
- **Velocity**—At what speed is new data generated?
- **Veracity** - refers to the trustworthiness and accuracy of the data.
- **Value** - emphasizes the importance of extracting meaningful insights from data.

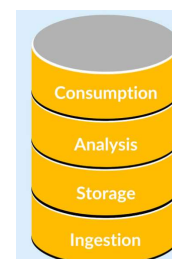


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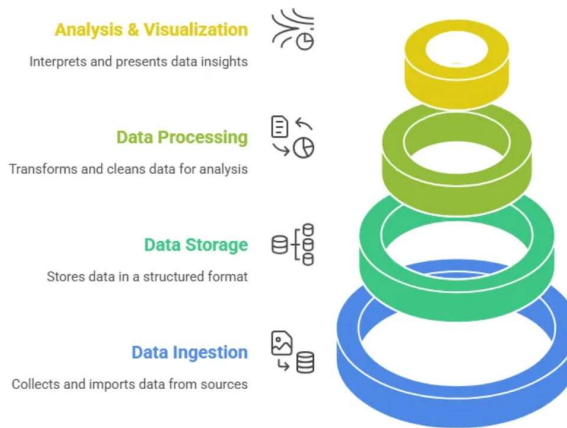
BIG DATA COMPONENTS

- **Ingestion** - Extract, transform and load (ETL) is the process of preparing data for analysis
- **Storage** - The final step of ETL is the loading process, where the converted data is stored in a data lake or warehouse and eventually processed.
- **Analysis** - In the analysis layer, data gets passed through several tools, shaping it into actionable insights. There are four types of analytics on big data: diagnostic, descriptive, predictive and prescriptive.
- **Consumption** - presenting the information in a format friendly and interpretable to the end-user. (forms of tables, advanced visualizations and even single numbers if requested)



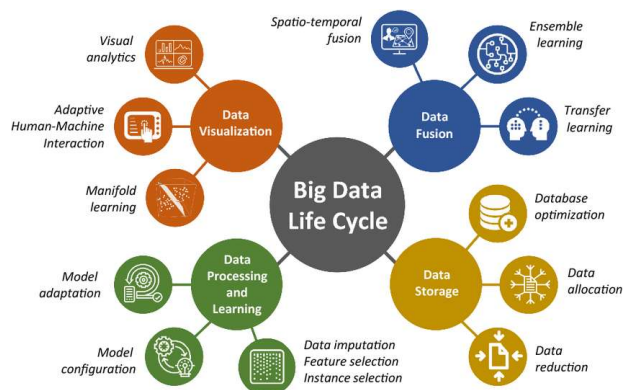
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BIG DATA ARCHITECTURE LAYERS



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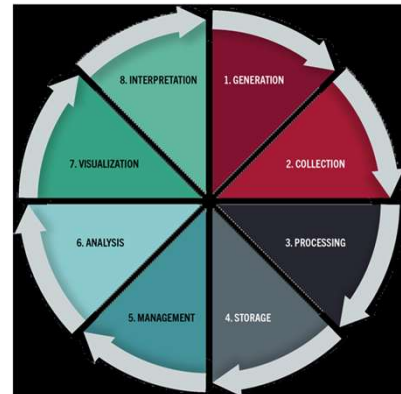
BIG DATA LYFE CYCLE



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BIG DATA LYFE CYCLE

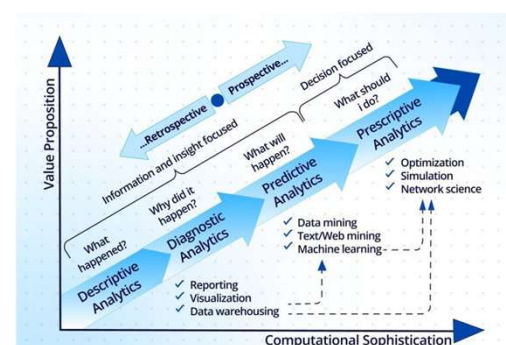
- **Generation** - everything that is learned, and knowledge extracted from the analysis of data, can generally be used for the next step.
- **Collection** – Forms; Surveys; Interviews; Direct observation.
- **Processing** - Data dispute (data is cleaned and is transformed into sets that are more accessible and useful); Data compression: data is transformed into a format that can be stored in a more efficient way; Data encryption (data are translated into a different code in order to be able to protect their privacy)
- **Storage** – creation of databases or data sets, that can be stored at cloud servers or physical storage servers. Establish security protocols and make security copies of all data that is going to be stored. A preventative measure in case the original source is corrupted or experiences some sort of compromise.
- **Management** - the management of databases or data sets that have been previously stored (organize, store and recover data as needed all across life cycle in a given project)
- **Analysis** – employ different tools and strategies (Statistical models, Algorithms, Artificial Intelligence, Data mining, Machine learning)
- **Visualization** - refers to the process of creating graphic representations of information, generally through the use of one or more visualization tools
- **Interpretation** - may not only include a description or explanation of what the data shows but, more importantly, what the implications may be



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TYPES OF BIG DATA ANALYTICS

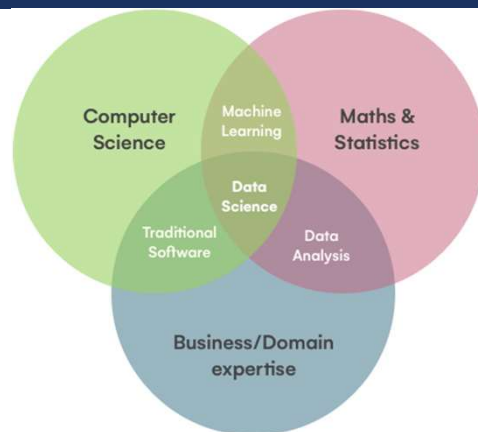
- **Descriptive analytics** answers the "what happened" question.
- **Predictive analytics** answers the "what is likely to happen" question.
- **Prescriptive analytics** solves the "how to make it happen" question.
- **Diagnostic analytics** solves the "why did it happen" question.



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DATA SCIENCE, BIG DATA, AND DATA ANALYTICS



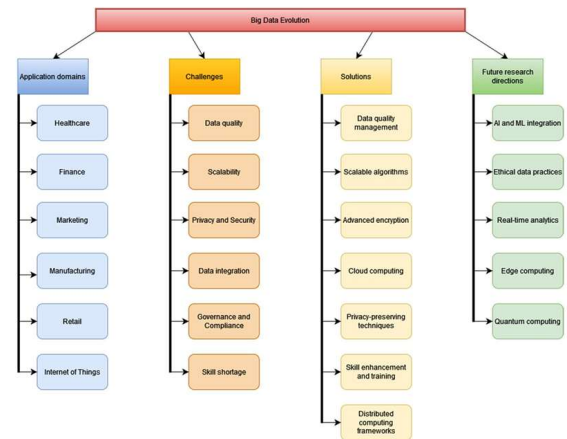
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BIG DATA CHALLENGES

- **Data storage** - Big data is ... BIG (volume, velocity, and variety)
- **Data security and privacy** needs will impact how IT teams manage that data
- Data is beneficial only if it is used
- **Big data technology is changing permanently ...**
 - Apache Hadoop was the popular technology
 - Apache Spark was introduced in 2014
 - Keeping up is an ongoing challenge - a combination of technologies are delivering new breakthroughs in the big data market.

CHALLENGES IN BIG DATA

- **Edge Computing:** Processing data closer to its source, at the edge of the network, to decrease latency and improve real-time decision-making.
- **AI and Machine Learning Integration:** Advanced AI and machine learning algorithms will drive more accurate predictions and automated decision-making.
- **Data Privacy and Security:** Stricter regulations and advanced security measures will address data privacy and protection concerns.
- **Hybrid and Multi-Cloud Solutions:** Organizations will leverage a combination of on-premises and cloud-based resources to efficiently manage and process big data.



Taxonomy of Big Data evolution over the last 15 years

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BIG DATA APPLICATIONS

- Retail and ecommerce
- Healthcare
- Financial services
- Manufacturing
- Government and public services

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